

Short Paper

What Holds an LLM-Generated Conversation Together? Manipulable Levers for Multi-Voice Literary Podcasts

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*What holds an LLM-generated multi-voice conversation together, and what can a designer manipulate to reshape it? We investigate these questions through a system that generates literary discussion podcasts about novels, identifying four levers that independently shape conversational coherence: (1) expert personas—stereotyped perspectives whose demand profiles control what each speaker seeks in the source material; (2) passage grounding—the selected passages that anchor quotation, prevent confabulation, and create material for cross-expert engagement; (3) conversational design instructions—prompt-level specifications for quote handling, reactive language, turn structure, and prosodic annotation; and (4) segment continuity—the mechanisms by which individually generated segments cohere into a single episode. We test the system on five novels—Dickens’s *Bleak House* and *Our Mutual Friend*, Eliot’s *The Mill on the Floss*, Gaskell’s *North and South*, and Forster’s *A Passage to India*—generating 196 multi-voice episodes across five passage selection conditions and multiple expert panel configurations. Key findings: expert personas dominate script character so strongly that different passage selection algorithms (transport vs. embedding; Jaccard 0.048) produce convergent scripts; passage grounding of any kind prevents confabulation (+69 percentage point quote verification advantage over ungrounded generation); and conversational design instructions produce measurable, stable effects—100% quote pattern compliance, 59 reactive markers per episode, stable expert airtime proportions across novels. Where the passage selection method does matter is in manipulability: formulating selection as min-cost flow makes every editorial choice visible as a cost and a constraint, enabling systematic exploration at zero marginal LLM cost. A planned prompt ablation study will isolate which conversational design instructions are necessary vs. emergent from the shared-passage architecture.*

1. Introduction

Google’s NotebookLM (Google 2024) popularised LLM-generated podcast conversations, but the conversational design behind such systems is largely unexamined. What makes a multi-voice LLM-generated discussion *cohere*—sound like a genuine conversation rather than parallel monologues? What levers does a designer have, and which ones actually matter?

We investigate these questions through a system that generates literary discussion podcasts about novels. The system exposes four independently manipulable levers:

1. **Expert personas.** Stereotyped perspectives (literary critic, social historian, close reader, etc.) with demand profiles that control what each speaker seeks in the source material. Swapping panels or adjusting demands reshapes the conversation’s intellectual character.
2. **Passage grounding.** The selected passages that anchor quotation, prevent confabulation, and create shared material for cross-expert engagement. We compare five selection methods including a min-cost flow formulation that makes every editorial choice visible as a cost and a constraint.
3. **Conversational design instructions.** Prompt-level specifications for quote handling, reactive language, turn structure, and utterance-level prosodic annotation. These are the instructions that tell the LLM to produce a *conversation*, not an essay.
4. **Segment continuity.** The mechanisms by which independently generated segments cohere into a single episode: bridging from the previous topic, previewing the next, avoiding redundant introductions.

We test the system on five novels spanning different authors, eras, and literary traditions: Dickens’s *Bleak House* (Dickens 1852–1853) and *Our Mutual Friend* (Dickens 1864–1865), Eliot’s *The Mill on the Floss* (Eliot 1860), Gaskell’s *North and South* (Gaskell 1854–1855), and Forster’s *A Passage to India* (Forster 1924), generating 196 multi-voice episodes across five passage selection conditions and multiple expert panel configurations.

The central finding is a *dominance hierarchy* among levers. Expert personas dominate script character so strongly that different passage selection algorithms (transport vs. embedding; passage Jaccard 0.048) produce convergent scripts. Passage grounding of any kind prevents confabulation (+69 pp quote verification). Conversational design instructions produce stable, measurable effects—but a planned ablation study (§6) will determine which are prompt-driven vs. emergent from the shared-passage architecture. Where the passage selection *method* matters is in manipulability: the min-cost flow formulation enables systematic editorial exploration at zero marginal LLM cost, a capability that embedding-based retrieval cannot match.

2. System Design

Figure 1 shows the four-phase pipeline. The enrichment layer (Phase 0) is computed once and shared by all downstream runs. Passage selection (Phases 1–2) varies by condition. Script generation (Phase 3) is identical across conditions.

2.1 Enrichment Layer

Each novel is split into passages (one per paragraph or small group of paragraphs) and each passage is annotated by an LLM (Haiku) with 20 structured fields: an interest score (0–5), seven provision dimensions (character_development, plot_advancement, thematic_depth, social_critique, humor_entertainment, atmosphere_setting, narrative_technique—each rated none/weak/strong as 0/1/2), char-

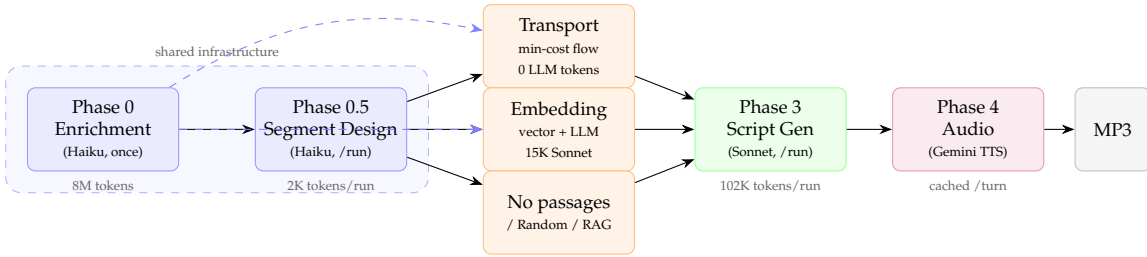


Figure 1
Pipeline architecture. The shared enrichment layer (blue, left) is computed once. Passage selection (orange, centre) varies by condition—transport uses zero LLM tokens. Script generation (green, right) is identical across conditions. Dashed arrows show enrichment data flowing to selection.

Table 1
Corpus statistics. All five novels are enriched with the same 20-field schema; passage counts reflect natural paragraph boundaries.

Novel	Chapters	Passages	Year
<i>Bleak House</i> (Dickens)	67	6,916	1853
<i>Our Mutual Friend</i> (Dickens)	63	5,775	1865
<i>The Mill on the Floss</i> (Eliot)	58	4,654	1860
<i>North and South</i> (Gaskell)	52	2,942	1855
<i>A Passage to India</i> (Forster)	37	2,556	1924
<i>Total</i>	277	22,843	

acters present, themes, emotional register, plot function, quotability, and a best_quote extraction.

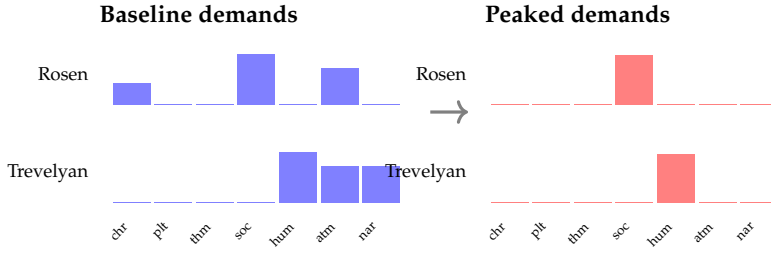
Table 1 shows the five novels and their passage counts. For the embedding pipeline, each passage also receives a contextual retrieval summary (Anthropic 2024) prepended before embedding with OpenAI `text-embedding-3-small`. Total enrichment cost across all five novels: ~25M Haiku tokens plus ~4.5M OpenAI embedding tokens, amortised across all runs at zero marginal cost.

2.2 Expert Personas and Demand Profiles

Six expert personas, three per episode: **Hartley** (literary critic), **Blackstone** (legal historian), **Woodcourt** (close reader), **Edmund** (traditionalist), **Rosen** (Marxist critic), **Trevelyan** (performer). Each has a *demand profile*: a vector over the seven provision dimensions. These are the primary editorial control surface (Figure 2).

2.3 Min-Cost Flow Formulation

Passage selection is a min-cost flow problem solved independently per provision dimension (Figure 3). Each eligible passage is a supply node whose supply equals its provision strength (strong = 2, weak = 1). Each expert with positive demand for the dimension is a demand node. A single null sink absorbs all supply that the solver does not route to experts, but at a cost equal to the passage’s interest score (0–5); discarding high-interest passages is expensive. The OR-Tools `SimpleMinCostFlow` solver requires

**Figure 2**

Demand profile manipulation. Left: baseline demands are spread across dimensions (e.g. Rosen: social = 14, atmosphere = 10, character = 6). Right: peaked demands concentrate each expert on their defining dimension. The transport solver responds by selecting qualitatively different passages.

exact balance (total supply = total demand + null sink demand) and raises an error otherwise.

Costs encode editorial preferences. On passage-to-cluster arcs, strong provisions cost 1 and weak cost 3; an interest penalty ($\max(0, 5 - \text{score})$) further discourages dull passages. Cluster-to-expert arcs are k parallel arcs of capacity 1 with costs $0, \lambda, 2\lambda, \dots$, creating a convex diversity penalty that discourages loading one expert with passages from the same chapter cluster. The solver minimises total cost, naturally preferring strong, interesting, diverse passages.

A passage is *selected* if it wins a competition in any dimension for any expert. The same passage may be assigned to multiple experts (each can discuss it). Results across all seven dimensions are merged; when the same passage–expert pair appears in multiple dimensions, the lowest-cost assignment is kept.

One flow is solved per provision dimension; results are merged into per-expert passage assignments via a survival rule: a passage is selected if it wins in any dimension for any expert. A second min-cost flow assigns selected passages to episode segments. Because solving a flow is cheap (<1 second, zero LLM tokens), the user can adjust demand profiles and re-solve interactively—the core of our manipulability hypothesis.

2.4 Five Selection Conditions

All conditions share the same segment design (Phase 0, Haiku) and script generation (Phase 3, Sonnet). They differ only in what passages are provided to Phase 3:

1. **No passages:** the LLM generates from prior knowledge alone.
2. **Random:** 32 passages sampled uniformly, assigned round-robin.
3. **Plain RAG:** top- k passages by cosine similarity between expert persona embeddings and raw passage text.
4. **Embedding + LLM curation:** contextual embeddings (Anthropic 2024) with Sonnet chain-of-thought curation ($\sim 15\text{K}$ tokens per run).
5. **Transport:** min-cost flow over enrichment features (0 LLM tokens for selection).

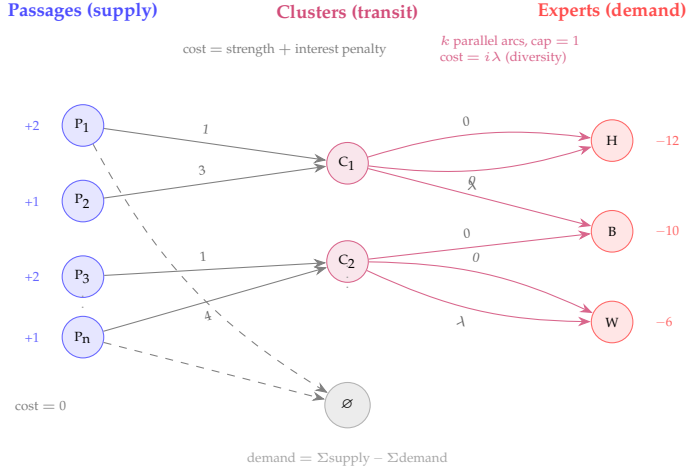


Figure 3

Per-dimension min-cost flow graph. Each eligible passage is a **supply node** (strong = 2, weak = 1 units). Each expert with demand for this dimension is a **demand node**. **Passage**→**cluster** arcs carry provision costs (strong = 1, weak = 3) plus an interest penalty ($\max(0, 5 - \text{score})$).

Cluster→**expert** arcs are k parallel arcs of capacity 1 with costs 0, λ , 2λ , ..., creating a convex penalty that discourages selecting multiple passages from the same chapter cluster. **Passage**→**null** arcs (dashed, cost = interest score) let the solver discard passages, but at a cost proportional to their interest rating; high-interest passages are therefore preferentially routed to experts rather than discarded. The null sink absorbs all supply not routed to experts. OR-Tools requires exact balance: total passage supply = total expert demand + null sink demand. The code verifies this and raises an error on any non-optimal status.

2.5 Audio Rendering (Phase 4)

The structured script is rendered to audio using Google’s Gemini 2.5 TTS models (gemini-2.5-flash-preview-tts or gemini-2.5-pro-preview-tts). The separation of script generation (Phase 3) from audio rendering (Phase 4) is deliberate: the structured output schema carries enough annotation to drive TTS without further LLM inference, and cached audio segments survive script iteration.

Each speaker is assigned a distinct Gemini voice (e.g., Sulafat for the host, Zephyr for Hartley, Sadaltager for Blackstone) with an accent direction embedded as a natural-language stage direction in the TTS prompt (“Home Counties accent, BBC Radio 4 style” for the host; “Edinburgh accent, dry and authoritative” for Blackstone).

The utterance-level annotations from Phase 3—rate multiplier, pause timing, emphasis words, and quote mode—are translated into natural-language prosody directions rather than SSML parameters:

- **Rate:** a multiplier of 0.93 becomes “Speak slowly and deliberately”; 1.04 becomes “Speak with brisk energy.”
- **Quote mode:** “setup” → “Build anticipation—this leads into a literary quotation”; “reading” → “Read this as a direct literary quotation with weight and relish.”

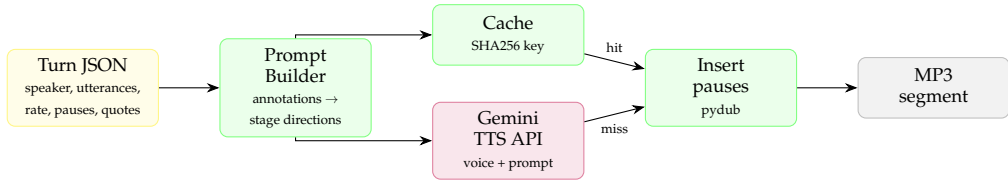


Figure 4

Audio rendering pipeline (Phase 4). Utterance annotations from Phase 3 are translated to natural-language stage directions. Rendered turns are cached by content hash; pauses are inserted in post-processing. Output is MP3 at 24 kHz mono.

- **Pauses:** `pause_before_ms` and `pause_after_ms` are realised as synthesised silence segments inserted during post-processing, not via TTS timing.
- **Emphasis:** up to 3 words per utterance, passed as “Give slight emphasis to: *Chancery, fog.*”

Figure 4 illustrates the rendering pipeline for a single turn.

Rendered audio is cached per turn (keyed by SHA-256 hash of model, voice, and prompt text), so re-rendering after script edits only re-synthesises changed turns. Final output is 24 kHz mono MP3. Phase 4 is not part of the experimental comparison—all five conditions use the same rendering pipeline—but the structured annotations it consumes are a design constraint on Phase 3’s output schema, and the natural-language prosody directions demonstrate that the utterance-level annotations (rate, quote mode, pauses) carry meaningful information beyond text.

The rendering pipeline is pluggable at the profile level: a `TTSPProfile` abstraction selects the model, voice mapping, pause scaling, and prompt builder. The results reported here use the *classic* profile (Gemini 2.5 TTS, short stage-direction prompts) uniformly across all conditions. As an alternative render of one panel, we have also produced a *trevelyan_v2* rendering of the Panel B *Bleak House* episode using `gemini-3.1-flash-tts-preview` and the six-strategy prompting scheme from Google’s speech-generation documentation—Audio Profile, Scene, Director’s Notes, Sample Context, Transcript, and audio-tag hints. The alternative profile is not part of the quantitative comparison; it is available on the demo site¹ and serves as a qualitative check that the Phase 3 output schema is stable under a different rendering backend.

3. Experimental Design

For *Bleak House*, 4 representative panels are run under transport (with updated demands and interest-based discard cost), while 20 panels from the original architecture are retained for embedding, no-passages, RAG, and random conditions: 84 BH episodes. For each of the other four novels, 4 panels are run under transport and 8 panels under embedding and no-passages: 80 cross-novel episodes. The total is 186 episodes across five novels. Across all conditions: identical segment design, identical script generation prompt, identical persona descriptions, identical Pydantic output schema. The only manipulation is what appears in the “Assigned Passages” section of the Phase 3 prompt.

¹ <https://bleakhouse-demo.fly.dev/examples>

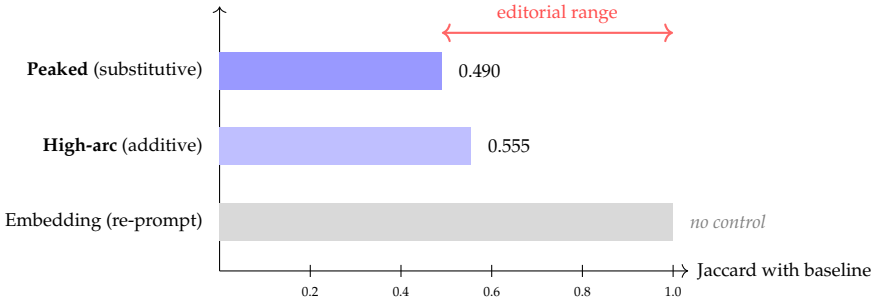


Figure 5

Manipulability comparison. Transport demand changes produce measurably different passage sets (Jaccard 0.490–0.555 with baseline). The embedding pipeline has no equivalent parameter; the user must re-prompt the LLM curator with no guarantee of systematic change.

Metrics. Character mention density (per 1,000 words, against each novel’s named character list), character entropy (Shannon 1948), unique characters per episode, quotes per episode, per-expert TF-IDF vocabulary signatures (cross-condition and cross-novel cosine), quote verification (fuzzy 5-word subsequence matching against source text), and material similarity (Jaccard, Jensen-Shannon divergence (Lin 1991), maximum mean discrepancy (Gretton et al. 2012)).

4. Results

4.1 Hypothesis 1: Manipulability

The central claim is that transport’s demand profiles provide a control surface that embedding-based retrieval lacks. We test this with two manipulation strategies applied to the transport pipeline (Figure 5).

Additive (high-arc). Increasing character arc demands adds passages proportionally to the extra demand units. Character mentions increase for the targeted characters, but the podcast is qualitatively unchanged. The manipulation is *predictable*: added passages trace directly to added demand units.

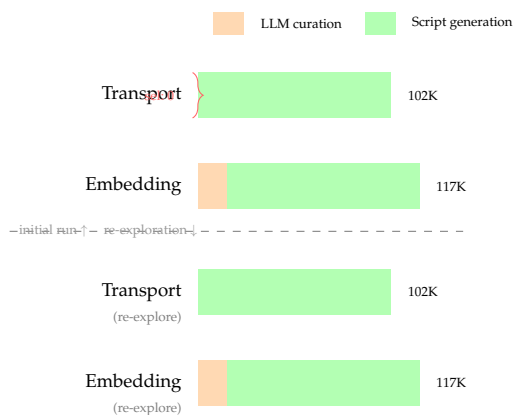
Substitutive (peaked). Peaking each expert’s demand vector—amplifying the defining dimension while zeroing secondaries—reshapes the passage set substantially. An expert’s passages become dominated by their peak dimension, while secondary dimensions are reallocated to other experts. The manipulation is *transformative*: it reshapes the editorial character of the episode through demand profile changes alone.

The embedding pipeline has no equivalent mechanism. The user cannot express “what if Rosen demands only social critique?” as a retrieval query. The only recourse is to rewrite the curation prompt and hope the LLM responds systematically.

4.2 Hypothesis 2: Exploration Cost

Figure 6 compares the token cost of exploring alternatives.

The transport solver reads enrichment features as numeric vectors—zero LLM tokens for selection. The embedding pipeline requires ~15K Sonnet tokens per run for LLM curation. Script generation (~102K tokens) dominates both, so the per-run saving

**Figure 6**

Token cost per run. Transport selection costs zero LLM tokens (red brace); embedding requires $\sim 15\text{K}$ Sonnet tokens for curation (orange). Script generation (green) dominates both. For re-exploration, transport re-solves instantly; embedding must re-prompt the curator. Over N explorations, transport saves $N \times 15\text{K}$ Sonnet tokens.

Table 2

Token budget by model class. Enrichment is computed once per novel; per-run costs are dominated by script generation.

Component	Haiku	Sonnet
Enrichment (once, 5 novels)	25M	—
Segment design (/run)	2K	—
Curation (/run, embed only)	—	15K
Script generation (/run)	—	102K
Project total (186 runs)	25.4M	19.0M

is modest (13%). But the cost advantage compounds with exploration: N alternative configurations cost $N \times 102\text{K}$ for transport vs $N \times 117\text{K}$ for embedding. More importantly, transport re-exploration requires only re-solving ($< 1\text{s}$) before committing to the expensive script generation step—the user can preview passage assignments, evaluate them, and decide whether to generate a script. The embedding pipeline offers no such preview.

Table 2 shows the full token budget.

The enrichment layer’s 25M Haiku tokens (across five novels) are amortised across all runs. Adding a new panel for one novel costs $\sim 300\text{K}$ Sonnet tokens (3 conditions $\times \sim 100\text{K}$ each).

4.3 Aggregate Quality Comparison

Table 3 shows that across all five novels, the selection conditions produce comparable scripts on surface metrics.

Embedding narrows: lowest character density (17.0/1Kw), fewest unique characters (10.6), lowest entropy (2.56). Transport produces the highest quote count (46.1/ep) and word count (10.3K), reflecting the interest-based discard cost which favours interesting

Table 3

Aggregate metrics across all 186 episodes. Transport (20 runs), embedding (54), RAG (20), no-passages (52), random (20).

Metric	Trns	Emb	RAG	NoPsg	Rnd
Words/ep	10.3K	9.7K	9.4K	9.7K	8.6K
Quotes/ep	46.1	45.6	37.7	22.9	33.6
Char/1Kw	20.1	17.0	22.9	23.6	21.5
Uniq chars	11.4	10.6	17.2	13.9	19.4
Char entropy	2.64	2.56	3.55	2.91	3.73

Table 4

Quote verification rates (fuzzy 5-word subsequence match). Top: *Bleak House* (5 conditions). Bottom: cross-novel (3 conditions).

Condition	Verified
<i>Bleak House:</i>	
Transport	99%
Embedding	97%
Plain RAG	95%
Random	94%
No passages	64%
<i>Cross-novel mean (3 conditions):</i>	
Transport	97%
Embedding	96%
No passages	20%

passages. Plain RAG matches transport on character density within 3%. No-passages produces the fewest quotes (22.9/ep)—less than half the grounded conditions—but achieves competitive character density from prior knowledge alone. These patterns confirm that selection method has modest impact on aggregate quality; the persona layer dominates.

4.4 Passage Grounding Prevents Confabulation

The gap between no-passages and any grounded condition dwarfs the gap between grounding methods. This holds across all five novels (Table 5): the cross-novel grounding gap averages +69 percentage points, ranging from +64 (Forster) to +89 (Dickens/OMF, Gaskell). Passage grounding prevents confabulation; the method of grounding is secondary.

Verification rates for grounded conditions are now uniformly high (94–99%), substantially improved from earlier runs. The no-passages condition varies widely across novels (9–64%), with *Bleak House* showing the highest rate, likely reflecting the LLM’s training exposure to this frequently studied text.

Table 5

Quote verification rates by novel (transport, embedding, no-passages). The grounding gap is large and consistent.

Novel	Trn	Emb	NP	Gap
<i>Bleak House</i>	99%	97%	64%	+35
Our Mutual Friend	95%	99%	10%	+89
Mill on the Floss	98%	97%	30%	+69
North and South	97%	95%	9%	+88
A Passage to India	96%	94%	31%	+64
<i>Mean</i>	97%	96%	29%	+69

Table 6

Material disjointness: transport $\leftrightarrow$ embedding, measured by passage and chapter Jaccard similarity. Near-zero passage overlap confirms the two methods select different material.

Novel	Pass. J	Ch. J	<i>n</i> panels
<i>Bleak House</i>	0.045	0.236	3
Our Mutual Friend	0.051	0.293	4
Mill on the Floss	0.062	0.307	4
North and South	0.027	0.351	4
A Passage to India	0.054	0.343	4
<i>Mean</i>	0.048	0.306	—

4.5 The Convergence Paradox: Personas Dominate Content

The strongest evidence for persona dominance is that transport and embedding select almost entirely different passages, yet scripts partially converge. Table 6 shows passage-level disjointness across all five novels.

Passage Jaccard is near-zero across all novels (mean 0.048): the two selection methods choose almost entirely different passages. Chapter Jaccard is higher (mean 0.306), indicating that while passages differ, the methods sometimes draw from the same chapters—natural given that both aim for the most interesting material. The convergence force is the expert persona: the same expert discussing different passages about the same novel arrives at overlapping vocabulary because the persona constrains what the expert *says about* the material. This is framework dominance: different inputs, similar outputs—because the persona, not the passage, dominates.

4.6 Expert Identity Is Robust

Expert personas maintain stable airtime signatures across novels (Table 7). Blackstone consistently commands the most airtime (23–26%), followed by Woodcourt (21–23%) and Hartley (17–21%). The alternative experts (Leigh, Rosen, Trevelyan) each hold 7–19% when present. Airtime proportions vary by at most 1.5 percentage points (SD) across the four non-BH novels, confirming that the persona prompt—not the source material—dominates each expert’s linguistic footprint.

Table 7

Expert airtime (% of total expert words, transport condition) across novels. Proportions are remarkably stable; the persona prompt dominates over source material.

Expert	OMF	MoF	NaS	PtI	SD
Blackstone	24.1	24.4	23.0	25.5	1.1
Woodcourt	20.9	21.3	22.9	22.9	1.0
Hartley	19.3	21.3	18.3	17.0	1.8
Leigh	18.6	16.9	18.2	17.0	0.8
Rosen	8.2	7.3	9.0	9.5	1.0
Trevelyan	9.0	8.8	8.7	8.1	0.4

Table 8

Conversational design measurements (186 episodes).

Feature	Measurement
Host turn fraction	22.7%
Quote pattern compliance	100.0%
TTS rate σ	0.027
Reactive markers/episode	59.0

4.7 Conversational Design Instructions Shape the Conversation

The script generation prompt encodes specific conversational design instructions—quote handling patterns, reactive language, turn structure, prosodic annotation—that go beyond content to shape *how* the conversation unfolds. These are the instructions that distinguish a discussion from parallel monologues. Across all 186 episodes:

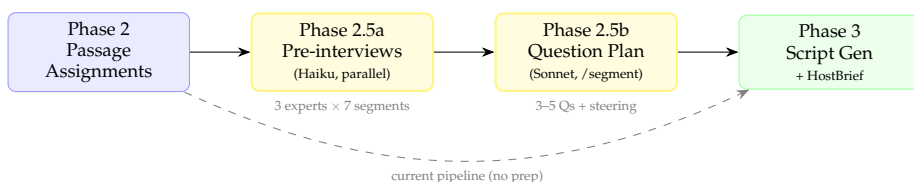
Quote pattern compliance is 100%: every quote utterance carries a passage reference or reading mode annotation. Reactive markers (59/ep) indicate genuine conversational engagement between experts. These results show that conversational design instructions produce their intended effects reliably. The open question—addressed by the ablation study (§6)—is which of these effects are *prompt-driven* (collapsing when the instruction is removed) vs. *architecture-driven* (emerging naturally from the shared-passage design regardless of prompting).

5. Planned: Host Preparation as a Conversational Lever

The weakest conversational metric in Table 8 is question frequency: 1.7 questions per segment. The host steers with statements and transitions rather than questions, producing discussion that flows but rarely probes. This exposes a fifth potential lever: *host preparation*—giving the host structured knowledge of what each expert will want to say before the conversation begins. We hypothesise that this will increase question frequency, improve cross-referencing, and produce more targeted conversational steering.

5.1 Design

Phase 2.5 inserts two sub-stages between passage assignment (Phase 2) and script generation (Phase 3), as shown in Figure 7.

**Figure 7**

Host preparation pipeline (Phase 2.5, planned). Pre-interviews (Haiku, parallel per expert per segment) feed into question planning (Sonnet), producing a HostBrief that is injected into Phase 3. Dashed arrow shows the current pipeline without preparation.

Phase 2.5a: Pre-interviews. For each expert–segment pair, the host (Haiku) conducts a short interview: “Given these passages, what strikes you? Where might you disagree with the other experts? Which passage would you most want to quote?” Each interview produces a `PreInterviewResponse` (key points, potential quotes, disagreement angles). Calls are parallel (~ 22 per run: $3 \text{ experts} \times 7.5 \text{ segments}$), costing $\sim 1\text{K}$ Haiku tokens each.

Phase 2.5b: Question planning. For each segment, the host (Sonnet) synthesises all three experts’ pre-interview responses and plans 3–5 questions designed to: draw out each expert’s strongest take, set up productive disagreements, create cross-referencing opportunities, and target specific passages for quotation. Output is a `HostBrief` (planned questions, steering notes, cross-engagement targets). One Sonnet call per segment ($\sim 3\text{K}$ tokens each).

Phase 3 integration. The `HostBrief` is appended to the Phase 3 user message alongside the passage assignments. The prompt instructs: “The host has prepared these questions—weave them naturally into conversation, not as rigid Q&A but as informed steering.”

5.2 Expected Cost

Phase 2.5a: $\sim 22\text{K}$ Haiku tokens per run (parallel, cheap). Phase 2.5b: $\sim 22\text{K}$ Sonnet tokens per run. Total: $\sim 44\text{K}$ additional tokens, a 43% increase over the current per-run budget but still dominated by Phase 3 script generation.

5.3 Hypotheses and Planned Evaluation

We will compare host-prepared vs unprepared episodes across at least $3 \text{ panels} \times 3 \text{ conditions}$ (9+ paired comparisons):

1. **Question frequency** increases from 1.7 to 3–5 per segment.
2. **Cross-expert referencing** increases beyond the current 61.4%, as questions are designed to create cross-engagement.
3. **Reactive language markers** increase, as experts respond to targeted questions rather than free-associating.
4. **Quote pattern compliance** is maintained or improves, as the `HostBrief` targets specific passages for quotation.

Table 9

Prompt ablation conditions (planned). Each condition removes one instruction from the Phase 3 system prompt. Baseline measurements are from 233 episodes with all instructions present.

ID	Instruction removed	Baseline metric	Hypothesis
A1	“Experts react to each other... not parallel monologues”	Expert→expert 68%, cross-refs 61.4%	Transitions become ho mediated; cross-refs
A2	Quote setup→reading→commentary pattern specification	Pattern compliance 74.9%	Quotes appear but wit consistent framing
A3	Rate multiplier and pause timing guidance	Rate $\sigma=0.027$, pause distributions	Values collapse to 1.0 or become erratic
A4	Sentence type classification requirement	8-type distribution (56% analysis)	Model defaults to “analysis” for most
A5	“At least one direct Dickens quote per expert turn”	8.1% quote_reading utterances	Quote frequency drop some turns ungroun
A6	Inter-speaker timing guidance table	pause_before mean 221ms	Pauses lose context- sensitivity
A7	Host steering instructions	Host fraction 23.5%	Host fraction changes unpredictably

5. **Expert identity** is not diluted—experts should still exhibit characteristic vocabulary despite more structured steering.

Status: designed, data models and pipeline integration planned, not yet implemented.

6. Planned: What Is Prompt-Driven vs. Architecture-Driven?

The conversational design measurements in Table 8 show that the Phase 3 prompt’s instructions produce their intended effects. But the central scientific question remains open: which conversational behaviours are *prompt-driven*—collapsing when the instruction is removed—and which are *architecture-driven*—emerging from the shared-passage design regardless of prompting? Does the model produce expert-to-expert transitions because the prompt says “this is a conversation, not parallel monologues,” or because the shared-context architecture (all experts see all passages) makes cross-engagement natural? Does the three-part quote pattern emerge from the explicit specification, or would the model structure quotes similarly without it?

6.1 Design

We identify seven distinct conversational instructions in the Phase 3 system prompt and ablate each one individually, keeping passage sharing and all other instructions intact. Table 9 lists the conditions and baseline measurements.

Each ablation is run on the default panel (Hartley, Blackstone, Woodcourt) with transport passages (v2 prompt), producing 7 ablated episodes compared against the existing baseline. Total cost: 7 runs $\times$ $\sim 102\text{K}$ Sonnet tokens = $\sim 714\text{K}$ Sonnet tokens.

6.2 Expected Findings

The ablation study will classify each instruction into one of three categories:

1. **Prompt-driven behaviours:** metrics that collapse when the instruction is removed. We expect the three-part quote pattern (A2) and rate/pause guidance (A3) to be prompt-driven—the model has no reason to produce these specific patterns without instruction.
2. **Architecture-driven behaviours:** metrics that survive ablation because the shared-context architecture produces them naturally. We hypothesise that cross-expert referencing (A1) may partially survive, since experts who see each other’s passages have natural reason to engage with them regardless of the prompt instruction.
3. **Redundant instructions:** metrics that are unaffected because the model’s default behaviour already matches the instruction. Sentence type classification (A4) may be redundant if the model naturally produces functional variety in literary discussion.

The most interesting potential finding is a dissociation between A1 and the cross-referencing metric: if removing “experts react to each other” does *not* reduce cross-referencing, then the shared-passage architecture—not the prompt instruction—is the active ingredient. This would strengthen the system design argument that passage sharing is the key architectural decision.

Status: ablation framework designed, 7 prompt variants and analysis pipeline planned, not yet implemented.

7. Discussion

7.1 A Dominance Hierarchy of Conversational Levers

The results reveal a clear hierarchy among the four levers that shape conversational coherence:

Personas dominate. Expert identity signatures are remarkably stable across novels, selection conditions, and panel configurations. The same persona discussing different passages about different novels produces overlapping vocabulary and consistent airtime proportions. This is the strongest force shaping what a conversation *sounds like*.

Grounding prevents confabulation. Any form of passage grounding (+69 pp quote verification) dwarfs the difference between grounding methods. The shared-passage architecture—all experts see all passages—may also be the active ingredient for cross-expert engagement, though the ablation study (§6) is needed to confirm this.

Conversational instructions produce measurable effects. Quote pattern compliance, reactive markers, and prosodic annotation are all prompt-driven. But which instructions are *necessary* vs. which are redundant given the architecture remains an open question—precisely the question the ablation study addresses.

Passage selection method matters for manipulability, not quality. Transport and embedding produce comparable scripts from almost entirely different passages. The transport formulation’s contribution is not better scripts but *editorial control*: the demand-profile mechanism allows systematic, predictable manipulation at zero marginal LLM cost.

7.2 Stereotypes as Editorial Objects

The expert personas are deliberately stereotypical. Following Boden (2004), the system performs combinational creativity—but the interesting creativity is the user’s, exploring the configuration space and confronting the stereotypes that make the machinery work. Making stereotypes explicit as demand vectors, rather than implicit in a prompt, turns editorial bias into a first-class object that can be inspected, compared, and argued about.

7.3 Limitations

All evaluation is automated; no human listeners have judged the generated episodes. Statistical comparisons are descriptive; bootstrap CIs would strengthen the claims. The manipulability advantage assumes a user who wants to explore configurations—for one-shot generation, the transport pipeline offers no quality advantage over plain RAG. The five-novel study uses the same automated metrics throughout; human evaluation remains future work. All five novels are from the English literary canon (1853–1924); the pipeline has not been tested on non-literary text, non-English sources, or contemporary works.

7.4 Implications

The central insight is that multi-voice LLM-generated conversations are shaped by a hierarchy of design levers, not a single algorithmic choice. Personas, grounding, conversational instructions, and segment continuity each contribute independently, and understanding which levers dominate—and which are redundant—is essential for principled system design. This framing applies beyond literary podcasts to any domain where LLMs generate structured multi-party discourse: legal panel discussions, curriculum dialogues, exhibit narration, or simulated expert consultations.

Acknowledgments

This work benefited from the infrastructure of LexisNexis Legal & Professional and conversations with colleagues at The Ohio State University. All experiments used the Anthropic and OpenAI APIs.

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